

Video Transcript

Video 3: Mapbox: A Web Client Library (03:55)

Next, let's take a look at the front end that the small web client that we have provided you. This is using a library, a great free library called 'mapbox'. and it's supported by the chrome browser. Now before I go any further, I should say that if you do not change anything on this client, this will still work. But I'm going to provide you with the background and walk through the code in case you want to make any changes or you want to do any, any extensions to what we have provided you. So, let's take a look at that code now.

As you can see here, I have the index.html file. This is the file that will be served if you hit the root of the application by flask. The top of the file brings in the mapping libraries from mapbox. It establishes some style and then the rest of this file defines the application. You can see here that you will need a token. In this point, I have simply written in there your token, but you can get your actual token from this address. I had mine before, but I updated it with your token simply to illustrate that you will need your own.

This is a great one to have, as mentioned it's free, and you can use these maps on any application you like. Now most of the application involves querying the server, that is asking for updated information from the server and then updating the location of the buses on the screen. Now you can see here that there are a couple of global variables. The first one is map, the second one is markers which will hold all of the buses locations in the visual marker on the screen. We then create a map and this is aided by that library and then we initialize the application by adding markers.

The very first thing that happens is that we get the bus locations. This calls the location on the server/location and brings back that json data. Then that is parsed and we return an object to this addMarkers function. The very first thing that we do after that is that then we loop through all of the buses that have been sent. If we find the marker for it, we simply update its location as you can see here, move marker. Otherwise we add a new marker to the screen. Now below this, there are all the functions to support that and you can read through those on your own.

Now let's fire up the server, bring up the web client, and take a look at that and see what it looks like. So, let's fire up the server. As you can see there, that's up and running. Let's open up the browser and the address is localhost:3000 and that's 3000. And as you can see there, this is monstrous. Let's go ahead and make the size on this a little bit more reasonable. Go ahead and move in here. And if we took a look at it for a little while, you see the move. You can see there that they just move.

The other thing to note as well is that they're not following precisely route one, because this is simply doing a linear mapping from one end of that bounding box to the other. But it's a pretty good one here to simply understand what's taking place. You're still working with the simulated server. If you wait for them to get all the way to the end and come back, you'll see them change color as well. So, this is a good one to get understanding on and then, once you do, you can go forward.

